

BHAVESH SINGH RATHOD

312-823-7445 | bhaveshsinghrathod@gmail.com | [LinkedIn](#) | [Github](#)

SUMMARY

CompTIA Security+ certified Cybersecurity Engineering graduate with hands on experience building AI-augmented SOAR pipelines using Wazuh SIEM, custom Python and PowerShell automation via REST APIs and multi-OS endpoints monitoring across Windows, macOS, Linux, iOS and Android using Wazuh EDR agents. Demonstrated ability to reduce mean alert triage time by over 90% through automated incident response workflows and AI assisted alert triage agents. Targeting SOC Analyst, Security Operations Analyst, Security Automation Engineer and SIEM Engineer roles with a focus on scalable detection engineering in an AI-native threat landscape.

EDUCATION

- Master of Cybersecurity Engineering 08/2024 - 05/2026
Illinois Institute of Technology, Chicago, IL
- Bachelor of Engineering, Computer Engineering 08/2020 - 06/2024
Savitribai Phule Pune University, Pune, Maharashtra, India

SKILLS

Security Operations: Wazuh SIEM, Wazuh EDR, SOAR Automation, Alert Triage & Escalation, Log Ingestion & Correlation, Incident Response, Intrusion Detection, Threat Intelligence Workflows, Vulnerability Management, Endpoint Detection & Response (EDR)

Detection & Analysis: Network Traffic Analysis, Cyber Forensics, Custom Detection Scripting, Back-testing Framework, MITRE ATT&CK Framework, NIST CSF

Endpoint & Infrastructure: Multi-OS Environments (Windows, macOS, Linux, iOS, Android), Kali Linux, Active Directory, VMWare / VirtualBox, Docker, Git/Github

Languages & Automation: Python, PowerShell, Bash, Go, JAVA, REST API Integration

Security Tools: Burp Suite, Wireshark, Nmap

Cryptography & Architecture: SHA-256, Hashing Algorithms, Cryptography, Blockchain Architecture, Network Security Protocols (TCP/IP)

CERTIFICATES

CompTIA Security+ | CompTIA | June 2026 – June 2029 | Credential ID: 833460f9513f40068c3145d87b0d6a21

Try Hack Me Advent of Cyber 2024 | Try Hack Me | December 31, 2024 | 24 Cybersecurity Challenges

Telstra Cybersecurity Simulation | Forage | July 27, 2024

NSE 1 Network Security Associate | Fortinet | March 2023

Goldman Sachs Cybersecurity Simulation | Forage | September 5, 2022

Mastercard Cybersecurity Simulation | Forage | July 10, 2024

Cryptography and Network Security | NPTEL / IIT Kharagpur | Jul-Oct 2022

PROJECTS

Enterprise Multi Model SOAR Automation

03/2026 - 05/2026

- Architected an end-to-end agentic SOAR pipeline powered by local LLM (Using OLLAMA) that ingest security logs, autonomously ranks the top 5 priority incidents by confidence score, and selects the optimal mitigation scripts from a curated response library.
- Implemented human-in-the-loop approval workflow where the AI presents its recommended mitigation action and confidence score, awaiting operator approval before execution – with fallback option to manually select an alternative response script or escalate.
- Automated incident response actions including alert suppression, IP blocking and endpoint isolation upon operator approval, mapped to MITRE ATT&CK tactics (Initial Access, Execution, Persistence, Lateral Movement) for structured threat categorization.
- Built a React-based frontend dashboard integrated via REST API to enable full operator control – log review, AI recommendation viewing, approval / rejection of mitigation actions, and manual script selection – reducing mean triage time from 45 minutes to under 90 seconds.
- Engineered automated executive report generation producing a 1page post incident summary of detected threats, confidence scores, actions taken and escalation decisions for stakeholder communication.

Wazuh Home Lab – Multi-OS Threat Detection & Log Analysis

12/2025 - 02/2026

- Built a local adversarial lab environment using VirtualBox on macOS, deploying Kali Linux as attack machine and Ubuntu (headless server) as victim endpoint with Wazuh EDR agent for real time log forwarding to a centralized Wazuh SIEM dashboard.
- Executed attacks simulations including brute force / credential attacks, Nmap port scanning and enumeration and malware behavior observation against the Ubuntu target, analyzing generated alerts and endpoint telemetry in the Wazuh dashboard to understand detection logic and logic correlation.
- Designed and tuned Wazuh detection rules and alert threshold across a 6-device multi-OS environment, replicating enterprise scale SOC monitoring and triage procedures.
- Conducted continuous vulnerability assessment workflows against the multi-OS environment, documenting findings and remediation steps aligned with MITRE ATT&CK framework techniques.

Blockchain-Based Document Storage and Authentication System, SPPU

11/2023 - 04/2024

- Led a 3-member team to design and deploy a blockchain-based document authentication system, successfully securing and authenticating multiple documents using a decentralized architecture.
- Replaced traditional centralized storage to eliminate single points of failure and reduce unauthorized access risk, implementing SHA-256 hashing for tamper-evident, cryptographically verifiable document retrieval on every request.

Image Encryption and Decryption, Independent

10/2023 - 11/2023

- Built a Python based image encryption tool supporting all major file formats, implementing symmetric encryption to preserve confidentiality of locally stored asset
- Designed an end to end encrypt-store-decrypt pipeline ensuring data remains unreadable, with full recovery on authenticated decryption request

EXPERIENCE

Intern

Whitebridge Capital, Singapore (remote)

01/2026 - 04/2026

- Monitored live network traffic and system logs across trading infrastructure in a high availability financial environment, identifying and escalating cross domain performance anomalies and policy violations for further review.
- Supported infrastructure stability by diagnosing and resolving system performance bottlenecks across trading systems, maintaining uptime requirements critical to real time trading operations.
- Managed version control repositories using Git to track configuration changes and ensure deployment stability across distributed remote infrastructure.